

Math Solutions

Introduction/ Common:

(9)

N=165	Predicted NO	Predicted YES
Actual NO	50	10
Actual YES	5	100

Here,

$$N = 165$$

$$TP = 100$$

$$TN = 50$$

$$FP = 10$$

$$FN = 5$$

$$\therefore \text{Accuracy} = \frac{TN + TP}{TN + FP + TP + FN} = \frac{50 + 100}{165}$$

$$= 0.90909 = 91\%$$

$$\therefore \text{Precision} = \frac{TP}{TP + FP} = \frac{100}{100 + 10}$$
$$= 0.90909 = 91\%$$

$$\therefore \text{Recall} = \frac{TP}{TP + FN} = \frac{100}{100 + 5} = 0.95238 = 95\%$$

$$\therefore \text{F1-score} = 2 * \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 90 \times 95}{90 + 95}$$
$$= 0.92956 = 93\%$$

$$\text{FN Rate} = \frac{FN}{TP + FN} = \frac{5}{100 + 5} = 0.0476 = 5\%$$

$$\text{TN Rate} = \frac{TN}{TN + FP} = \frac{50}{50 + 10} = 0.8333 = 83.33\%$$

Q) Given,

13, 15, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25,
30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Here total values, $n = 26$

a) Arithmetic mean, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

$$= \frac{1}{26} \times 793$$
$$= 30.5$$

median = $[(n+1)/2]^{\text{th}}$ observation if, $n = \text{odd}$

mean of $(n/2)^{\text{th}}$ observation and

$[(n/2)+1]^{\text{th}}$ observation, if $n = \text{even}$

here, $n = 26$ which is even,

$n/2^{\text{th}}$ observation = 25

$(n/2)+1^{\text{th}}$ = 30

$$\therefore \text{Median} = \frac{25+30}{2} = 27.5$$

b) Mode for ungrouped data can be simply find by counting their frequency

Value	13	15	16	19	20	21	22	25	30	33	35	36	40	45	46	50	70
Count	1	1	1	1	2	1	2	4	1	2	4	1	1	1	1	1	1

25 and 35 appears 4 times. They are modal values

comment: Here 2 values are mode. So Bimodal

c) Midrange = $\frac{\text{Max}(x) + \text{min}(x)}{2} = \frac{70 + 13}{2} = 41.5$

Association Rule:

Given,

TID	Item Sets
T1	A, B
T2	B, D
T3	B, C
T4	A, B, D
T5	A, C
T6	B, C
T7	A, C
T8	A, B, C, E
T9	A, B, C

~~A = 1, 2, 3, 4, 5~~
~~B = 1, 2, 3, 4~~

Minimum support = 2
 Minimum confidence = 50%

Step-1 Calculating Candidate Set (C1) and Frequent Items (L1)

C1

Item	count
A	6
B	7
C	5
D	2
E	1
F	

L1 (minimum Support)

item	count
A	6
B	7
C	5
D	2

∴ A, B, C, D

E = 2 is not frequent

Step-2 Generation C2, and L2:

C2 using L1 (pair of items) from main table →

Item set	count
{A, B}	4
{A, C}	4
{A, D}	1
{B, C}	4
{B, D}	2
C, D	0



L2:

Itemset	count
{A, B}	4
{A, C}	4
{B, C}	4
{B, D}	2

∴ A, B, C, D

Step-3 Generation C3, L3

$C3 \leftarrow C2 \cup A$ but $A \leftarrow C2 \cup B$ is not possible

Item	count
$\{A, B, C\}$	2
$\{A, B, D\}$	0
$\{A, C, D\}$	0
$\{B, C, D\}$	1

$\therefore L3 \rightarrow \{A, B, C\}$

(only one subset)

Step-4: Finding Association Rule for subset

$\text{support}(A \rightarrow B) = \text{support-count}(A \cup B)$

$\text{confidence}(A \rightarrow B) = \frac{\text{support-count}(A \cup B)}{\text{support-count}(A)}$

Rules	support	confidence
$A \wedge B \rightarrow C$	2	$\frac{\text{sup}\{A \cup B \cup C\}}{\text{sup}(A \cup B)} = 2/4 = 50\%$
$B \wedge C \rightarrow A$	2	$\frac{\text{sup}\{B \cup C \cup A\}}{\text{sup}(B \cup C)} = 2/4 = 50\%$
$A \wedge C \rightarrow B$	2	$\frac{\text{sup}\{A \cup C \cup B\}}{\text{sup}(A \cup C)} = 2/4 = 50\%$
$C \rightarrow A \wedge B$	2	$\frac{\text{sup}\{C \cup (A \cup B)\}}{\text{sup}(C)} = 2/5 = 40\%$
$A \rightarrow B \wedge C$	2	$\frac{\text{sup}\{A \cup (B \cup C)\}}{\text{sup}(A)} = 2/6 = 33.33\%$
$B \rightarrow B \wedge C$	2	$\frac{\text{sup}\{B \cup (B \cup C)\}}{\text{sup}(B)} = 2/7 = 28.1\%$

As minimum confidence = 50%

$\therefore A \wedge B \rightarrow C$, $B \wedge C \rightarrow A$, and $A \wedge C \rightarrow B$

are strong association rules.

PCA:

Given data points = $\{2, 3, 4, 5, 6, 7; 1, 5, 3, 6, 7, 8\}$

making them, 2D = $(2, 1), (3, 5), (4, 3),$
 $(5, 6), (6, 7), (7, 8)$

a)

Step-01: feature vector

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Step-02: mean vector (μ)

$$x = (2+3+4+5+6+7)/6 = 4.5$$

$$y = (1+5+3+6+7+8)/6 = 5$$

$$\therefore \text{mean vector } (\mu) = \begin{bmatrix} 4.5 \\ 5 \end{bmatrix}$$

Step-03: Subtract mean vector from feature vector

$$X_1 - \mu = (2 - 4.5, 1 - 5) = (-2.5, -4)$$

$$X_2 - \mu = (3 - 4.5, 5 - 5) = (-1.5, 0)$$

$$X_3 - \mu = (4 - 4.5, 3 - 5) = (-0.5, -2)$$

$$X_4 - \mu = (5 - 4.5, 6 - 5) = (0.5, 1)$$

$$X_5 - \mu = (6 - 4.5, 7 - 5) = (1.5, 2)$$

$$X_6 - \mu = (7 - 4.5, 8 - 5) = (2.5, 3)$$

$$\begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 2.5 \\ 3 \end{bmatrix}$$

Step-04: Calculate covariance matrix

$$\text{covariance matrix} = \frac{1}{n} \sum (X_i - \mu)(X_i - \mu)^t$$

$$\therefore m_1 = \frac{1}{6} \begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -2.5 & -4 \end{bmatrix} = \begin{bmatrix} 6.25 & 10 \\ 10 & 16 \end{bmatrix}$$

$$\therefore m_2 = \frac{1}{6} \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -1.5 & 0 \end{bmatrix} = \begin{bmatrix} 2.25 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore m_3 = \frac{1}{6} \begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} -0.5 & -2 \end{bmatrix} = \begin{bmatrix} 0.25 & 1 \\ 1 & 4 \end{bmatrix}$$

$$0 = 0.8 + 1 \times 2.8 = \frac{1}{6} \times 1.8$$

$$m_4 = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 0.25 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$m_5 = \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 1.5 & 2 \end{bmatrix} = \begin{bmatrix} 2.25 & 3 \\ 3 & 4 \end{bmatrix}$$

$$m_6 = \begin{bmatrix} 2.5 \\ 3 \end{bmatrix} \begin{bmatrix} 2.5 & 3 \end{bmatrix} = \begin{bmatrix} 6.25 & 7.5 \\ 7.5 & 9 \end{bmatrix}$$

$$\therefore \text{Covariance matrix} = \frac{1}{6} \begin{bmatrix} 17.5 & 22 \\ 22 & 39 \end{bmatrix}$$

$$\begin{bmatrix} 2.9 \\ 2.1 \end{bmatrix} \begin{bmatrix} 2.0 \end{bmatrix} \begin{bmatrix} 2.0 \end{bmatrix} \begin{bmatrix} 2.1 \end{bmatrix} \begin{bmatrix} 2.1 \end{bmatrix} = \begin{bmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{bmatrix}$$

Step-5: calculate eigen values

$$|m - \lambda I| = 0$$

$$\begin{vmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{vmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2.92 - \lambda & 3.67 \\ 3.67 & 5.67 - \lambda \end{vmatrix} = 0$$

$$\therefore \lambda^2 - 8.59\lambda + 3.09 = 0$$

$$\therefore \lambda = 8.22, 0.38 \quad \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ corresponding to}$$

Eigen value (greatest) is the principle component for the given dataset.

$$\therefore M X = \lambda X$$

M = covariance matrix

X = Eigen vector

λ = Eigen value

$$\begin{bmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 8.22 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\therefore \text{or, } 2.92x_1 + 3.67x_2 = 8.22x_1$$

$$\text{or, } 5.3x_1 = 3.67x_2 \quad \dots \text{--- (1)}$$

$$\therefore 3.67x_1 + 5.67x_2 = 8.22x_2$$

$$\text{or, } 3.67x_1 = 2.55x_2$$

$$\therefore \boxed{x_1 = 0.69x_2} \quad \therefore \text{P.C} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2.55 \\ 3.67 \end{bmatrix}$$

$$= |2.92 - 8.22| + |3.67 - 0| + |(0-1)| + |(0.38 - 5.67)| =$$

$$12 = 5 + 2 + 1 + 2 =$$

b) given $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $88.0, 88.8 = \mu$

Transform = Transpose of Eigen vector *
(Feature vector - mean vector)

$$= \begin{bmatrix} 2.55 \\ 3.67 \end{bmatrix}^T \times \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 4.5 \\ 5 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 2.55 & 3.67 \end{bmatrix} \times \begin{bmatrix} -2.5 \\ -4 \end{bmatrix}$$

$$= -21.055$$

K-means:

③ given $(22, 1, 42, 20)$ and $(20, 0, 36, 8)$

a) Manhattan distance, $d_m = \sum_{i=1}^n |p_i - q_i|$

$$= |(22-20)| + |(1-0)| + |42-36| + |20-8|$$

$$= 2 + 1 + 6 + 12 = 21$$

b) minkowski distance, $d_{mk} = \left(\sum_{i=1}^n |p_i - q_i|^m \right)^{1/m}$
here, $m=4$,

$$\therefore d = \left((122-201)^4 + (11-01)^4 + (142-361)^4 + (120-81)^4 \right)^{1/4}$$

$$= \left(2^4 + 1^4 + 6^4 + 12^4 \right)^{1/4}$$

$$= \left(16 + 1 + 1296 + 20736 \right)^{1/4}$$

$$= (22049)^{1/4}$$

$$= 12.19$$